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# DETF-02

## B. TECH. (FIFTH SEMESTER)

## MID SEMESTER EXAMINATION, Oct., 2022

### POWER PLANT ENGINEERING

**Time : 1½ Hours**

**Maximum Marks : 50**

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) What are the different types of power plant where electricity is produced in bulk quantities ? (CO1)

OR

(b) A power plant has the following annual factors : (CO1)

Load factor = 70%, Capacity factor = 50%, Use factor = 60%, Maximum demand = 20 MW.

Estimate :

(i) Annual energy production

(ii) Reserve capacity over and above the peak load

2. (a) Define Depreciation. What do you mean by sinking-fund method of accumulating the money for the depreciation fund ? (CO1)

OR

(b) A thermal power plant of 210 MW capacity has the maximum load of 160 MW. Its Annual load factor is 0.6. The coal consumption is 1 kg per kWh of energy generated and the cost of the coal is ₹ 500 per tonne. Calculate the annual revenue earned if energy is sold at ₹ 2 per kWh and the capacity factor of the plant. (CO2)

3. (a) Explain the factors to be considered while selecting the site for steam power plant. (CO2)

OR

(b) Steam with absolute velocity of 350 m/s is supplied through a nozzle to a single stage impulse turbine. The nozzle angle is 15°. The mean diameter of the blade rotor is 2 metre and it has a speed of 4000 r.p.m. Find suitable blade angles for zero axial thrust. If the blade velocity co-efficient is 0.9 and the steam flow rate is 10 kg/s, calculate the power developed. (CO2)

4. (a) Give the layout of a modern steam power plant. Explain all the circuits. (CO2)

OR

(b) Describe the various methods of steam turbine governing. (CO2)

5. (a) What are Supercritical Boilers ? Explain any *one* in detail. (CO2)

OR

(b) Discuss various Pressure velocity compounding in steam turbines.(CO2)